The three-equation melt condition at an ice–water interface

and its form in `solver2d`

1 Geometry and notation

The ice occupies $y < 0$ and the water $0 < y < L_y$; the ice–water interface is the wall $y = 0$ (in `solver2d`: `bc_bot = 'noslip'`, `bc_top = 'freeslip'`, gravity along $+y$). The unit normal $\mathbf{n} = \mathbf{e}_y$ points from the interface into the water. The subscript b denotes values at the interface ($y = 0$). The melt (ablation) rate m is the speed at which the interface recedes into the ice (in $\mathrm{m\,s^{-1}}$ of ice), positive for melting. The effect of pressure on the freezing point is neglected throughout (it would add a term $\lambda_3 p_b$, with $\lambda_3 \approx -7.5 \times 10^{-4}\,^\circ\mathrm{C\,dbar^{-1}}$, to (1)).

symbol	meaning	typical value
$T,\ S$	water temperature, salinity	
$T_b,\ S_b$	temperature and salinity at the interface	unknown
m	melt rate (ice thickness per unit time)	unknown
λ_1	liquidus slope	$\approx -5.7 \times 10^{-2}\ ^\circ\mathrm{C\,psu^{-1}}$
λ_2	liquidus offset	$\approx 8.3 \times 10^{-2}\ ^\circ\mathrm{C}$
L_f	latent heat of fusion	$3.34 \times 10^5\ \mathrm{J\,kg^{-1}}$
$c_w,\ c_i$	specific heat of water, ice	$\approx 3974,\ 2009\ \mathrm{J\,kg^{-1}K^{-1}}$
$\rho_w,\ \rho_i$	density of water, ice	$\approx 1028,\ 917\ \mathrm{kg\,m^{-3}}$
$\kappa_T,\ \kappa_S$	molecular diffusivities of heat and salt in water	$\approx 1.4 \times 10^{-7},\ \approx 7 \times 10^{-10}\ \mathrm{m^2 s^{-1}}$
κ_i	thermal diffusivity of ice	$\approx 1.1 \times 10^{-6}\ \mathrm{m^2 s^{-1}}$
S_i	salinity of the ice	≈ 0

The liquidus constants differ slightly between sources; the values above are representative of those used in ice–ocean models (e.g. Holland & Jenkins 1999). Check them, and the other material properties, against the source you use.

2 The three equations

1. Freezing point (liquidus). The interface is at the local freezing temperature:

$$T_b = \lambda_1 S_b + \lambda_2. \tag{1}$$

2. Heat balance. The heat conducted from the water to the interface melts ice and is partly conducted into the ice:

$$\rho_w c_w \kappa_T \left.\frac{\partial T}{\partial y}\right|_b = \rho_i L_f\, m \ +\ \rho_i c_i \kappa_i \left.\frac{\partial T_i}{\partial y}\right|_b . \tag{2}$$

The last term (conduction into the ice, T_i the ice temperature) is often neglected, or modelled as $\rho_i c_i \kappa_i (T_b - T_s)/h_i$ with T_s the temperature at the far side of an ice layer of thickness h_i.

3. Salt balance. The ice holds (almost) no salt. The salt that the meltwater dilutes at the interface must be supplied by diffusion in the water:

$$\rho_w \kappa_S \left.\frac{\partial S}{\partial y}\right|_b = \rho_i \, m \, (S_b - S_i) \;\approx\; \rho_i \, m \, S_b. \tag{3}$$

Equations (1)–(3) determine the three unknowns T_b, S_b and m. For melting ($m > 0$) the interface is fresher than the water ($\partial S/\partial y|_b > 0$) and warmer than the freezing point of the far-field water. Since $\kappa_T/\kappa_S \approx 10^2$, salt diffusion limits the melt rate together with the heat supply.

Signs. With $\mathbf{n} = \mathbf{e}_y$ pointing into the water, the conductive fluxes *towards* the interface are $\rho_w c_w \kappa_T \, \partial T/\partial y|_b$ and $\rho_w \kappa_S \, \partial S/\partial y|_b$. This holds for the set-up above: ice at $y = 0$, water at $y > 0$, gravity along $+y$ (the ice is physically on top). If instead the ice is at $y = L_y$ (water at $y < L_y$, normal $-\mathbf{e}_y$), replace $\partial/\partial y$ by $-\partial/\partial y$.

Wall-normal velocity at the interface. As in Wells & Worster (2011), we use the reference frame in which the interface is fixed at $y = 0$ (it is quasi-planar and recedes slowly). In this frame the ice moves towards the interface at speed m, and melts there. Mass conservation across the interface, $\rho_i \, m = \rho_w \, v(0)$, gives the boundary condition

$$v(0) = \frac{\rho_i}{\rho_w} \, m \;>\; 0, \qquad u(0) = 0. \tag{4}$$

The velocity at the interface is positive: meltwater flows from the interface into the water. It is the ablation rate times ρ_i/ρ_w (≈ 0.89 for sea water, ≈ 0.92 for fresh water), i.e. it equals the volume flux of meltwater per unit area. It combines the advection of the water past the receding interface and the expansion of the ice on melting. This is condition (3.5a) of Wells & Worster (2011), $v = -(\rho_s/\rho_l)\,\dot{a}$ with $\dot{a} = -m$. (In the frame of the ice the water velocity at the interface is $v(0) - m = (\rho_i/\rho_w - 1)\, m < 0$: the volume contracts on melting.)

The interface is therefore *not* impermeable. The heat and salt balances (2) and (3) nevertheless contain only the diffusive fluxes on the water side, in either frame. In the interface-fixed frame the salt balance reads $\rho_i \, m \, S_i = \rho_w v(0) \, S_b - \rho_w \kappa_S \, \partial S/\partial y|_b$ (salt carried in by the ice = salt carried away by advection and diffusion); with (4) this is exactly (3). The same holds for heat.

3 Special cases

Stefan condition (fresh water). Without salt, (3) drops out and (1) becomes $T_b = T_m$ ($0\,^\circ\mathrm{C}$):

$$T_b = T_m, \qquad \rho_i L_f \, m = \rho_w c_w \kappa_T \left.\frac{\partial T}{\partial y}\right|_b - \rho_i c_i \kappa_i \left.\frac{\partial T_i}{\partial y}\right|_b. \tag{5}$$

Transfer-coefficient form (unresolved boundary layers). Ocean models do not resolve the diffusive sublayers and write the fluxes with transfer coefficients γ_T, γ_S and the friction velocity u_*:

$$\kappa_T \left.\frac{\partial T}{\partial y}\right|_b \to \gamma_T u_* (T_w - T_b), \qquad \kappa_S \left.\frac{\partial S}{\partial y}\right|_b \to \gamma_S u_* (S_w - S_b), \tag{6}$$

with T_w, S_w the values in the water outside the sublayer. Neglecting conduction into the ice and with $S_i = 0$, eliminating m and T_b gives a quadratic for S_b:

$$a \, S_b^2 + b \, S_b + c = 0, \qquad \begin{aligned} a &= c_w \gamma_T \lambda_1, \\ b &= -\left[c_w \gamma_T (T_w - \lambda_2) + L_f \gamma_S\right], \\ c &= L_f \gamma_S \, S_w, \end{aligned} \tag{7}$$

Since $\lambda_1 < 0$, $a < 0$ while $c > 0$, so the two roots have opposite signs and the physical root is the positive one (it satisfies $0 < S_b \le S_w$ for melting). Then T_b follows from (1) and $m = (\rho_w/\rho_i)\,\gamma_S u_*(S_w - S_b)/S_b$.

In a resolved simulation the gradients are computed directly at the wall and no transfer coefficients are needed. If the wall gradient is linearised as $\partial T/\partial y|_b \approx (T_1 - T_b)/\delta_T$ (and similarly for S), the same quadratic (7) follows with $\gamma_T u_* \to \kappa_T/\delta_T$, $\gamma_S u_* \to \kappa_S/\delta_S$, $T_w \to T_1$ and $S_w \to S_1$.

4 Non-dimensional form in `solver2d`

`solver2d` uses a length unit ℓ, a velocity unit U, and $Re = U\ell/\nu$. The diffusivities are $\kappa_T = U\ell/(Re\,Pr)$ and $\kappa_S = U\ell/(Re\,Sc)$. Write the scalars as

$$T = T_{\mathrm{ref}} + \Delta T\,\theta, \qquad S = S_{\mathrm{ref}} + \Delta S\,s, \tag{8}$$

(θ and s are scalars 1 and 2 of the code) and the melt rate as a water velocity in units of U:

$$m^* = \frac{\rho_i\, m}{\rho_w\, U}. \tag{9}$$

With y in units of ℓ, equations (1)–(3) become

$$\theta_b = \Lambda\, s_b + \theta_f, \qquad \Lambda = \frac{\lambda_1 \Delta S}{\Delta T}, \quad \theta_f = \frac{\lambda_1 S_{\mathrm{ref}} + \lambda_2 - T_{\mathrm{ref}}}{\Delta T}, \tag{10}$$

$$m^* = St\left(\frac{1}{Re\,Pr}\left.\frac{\partial\theta}{\partial y}\right|_b - q_i\right), \qquad St = \frac{c_w \Delta T}{L_f}, \tag{11}$$

$$\frac{1}{Re\,Sc}\left.\frac{\partial s}{\partial y}\right|_b = m^*\,(s_b + \sigma), \qquad \sigma = \frac{S_{\mathrm{ref}} - S_i}{\Delta S}, \tag{12}$$

with the Stefan number St and the non-dimensional conduction into the ice

$$q_i = \frac{\rho_i c_i \kappa_i}{\rho_w c_w U \Delta T}\left.\frac{\partial T_i}{\partial y}\right|_b \qquad (q_i = 0 \text{ if conduction into the ice is neglected}). \tag{13}$$

Equation (12) uses $S_b - S_i = \Delta S\,(s_b + \sigma)$. If S_{ref} is chosen as the ice salinity ($S_{\mathrm{ref}} = S_i$, e.g. 0), then $\sigma = 0$.

The non-dimensional melt rate is exactly the non-dimensional wall-normal velocity at the interface, (4):

$$\frac{v(0)}{U} = m^* > 0. \tag{14}$$

The new non-dimensional parameters are St, Λ, θ_f, σ (and q_i). In `solver2d` the buoyancy enters through the Richardson numbers `ri(1)`, `ri(2)` of θ and s, and the diffusivities through `pr(1)` $= Pr$ and `pr(2)` $= Sc$.

5 Implementation in `solver2d`

The model is switched on with `melt = .true.` in `&melt_model` (source: `src/melt.f90`). The parameters are St (`melt_st`), Λ (`melt_lambda`), θ_f (`melt_thf`) and σ (`melt_sigma`). The scalars `melt_t` and `melt_s` are θ and s; with `melt_s = 0` the Stefan condition is used. Heat conduction into the ice is neglected ($q_i = 0$).

Interface values as wall values. The conditions hold at every point x of the interface, and are nonlinear ($m^* s_b$) and coupled (θ and s). $\theta_b(x)$ and $s_b(x)$ are imposed as Dirichlet values on all Fourier modes of the (batched) scalar solves of each Runge–Kutta substep.

Implicit coupling (`melt_xdiff = 'implicit'`). The conditions must hold with the wall gradients of the *new* scalars. Lagging the interior values by one substep is unstable when diffusion across the first grid cell within a time step is fast, $\kappa\,\Delta t/\Delta y_1^2 \gtrsim 1$. The theta-method solve is linear in its wall value w: for every Fourier mode k the new wall gradient is

$$g_k = g_k^0 + R_k\, w_k, \tag{15}$$

where $R_k < 0$ is the response of the wall gradient to a unit wall value. R_k is computed with the solver itself (zero right-hand side, unit wall value), for θ and s separately, and cached per substep and Δt. The coupled problem is solved by the iteration

1. solve all scalars with the current wall values $w = (\theta_b, s_b)$ and evaluate the wall gradients g (first row of the compact D_1);
2. form $G = g - R_{\mathrm{ref}}\, w$ with a constant R_{ref} per scalar, and solve at every x the local conditions with the wall gradient $G + R_{\mathrm{ref}}\, w^{\mathrm{new}}$;
3. repeat until $\max|w^{\mathrm{new}} - w| < $ `melt_tol`.

At convergence the conditions hold exactly with the new gradients (R_{ref} cancels). For mode k the error is multiplied by at most $|1 - R_k/R_{\mathrm{ref}}|$, which is below 1 for $R_{\mathrm{ref}} = (\min_k R_k + \max_k R_k)/2$ whatever the time step.

Exact coupling without iteration (default). The iteration is needed only because R_k depends on k, through the implicit x-diffusion term $c\,k^2$ of the θ-method, while the nonlinear local conditions couple the Fourier modes. With `melt_xdiff = 'explicit'` the x-diffusion $-\kappa k^2 s$ of θ and s is advanced explicitly with the RK3 weights, and only the y-diffusion is implicit. The y-operator $1 - c\,D_2$ is then the same for every mode, so the response R and the unit response profile $h(y)$ ($(1 - cD_2)h = 0$, $h(0) = 1$, homogeneous condition at $y = L_y$) do not depend on k, and in physical space

$$g(x) = g^0(x) + R\, w(x) \tag{16}$$

holds at every x. The condition is then imposed exactly in one pass per substep: (i) solve with zero wall values, which gives s^0 and g^0; (ii) solve the local problem below with $a_T = R_\theta$, $a_S = R_s$ and $G = g^0$; (iii) set $s = s^0 + w(x)\,h(y)$, i.e. $\hat{s}_k = \hat{s}_k^0 + \hat{w}_k\,h$ for every mode. The cost is one scalar solve per substep instead of one per iteration. The explicit x-diffusion adds the RK3 stability limit $\Delta t \le 2.5/(\kappa\, k_{x,\max}^2)$ (with the safety factor `cfl_visc`); since Δx is much larger than the near-wall Δy, this is usually much weaker than the limits that apply anyway.

Local problem. With a_T, a_S the values of R_{ref} for θ and s, $z = s_b + \sigma$ and $G_z = G_s - a_S\sigma$, equations (10)–(12) reduce at every x to

$$B z^2 + (A - \kappa_S a_S)\, z - \kappa_S G_z = 0, \qquad A = St\,\kappa_T\,[G_\theta + a_T(\theta_f - \Lambda\sigma)]\,, \quad B = St\,\kappa_T\,a_T\Lambda \ge 0, \tag{17}$$

with $\kappa_T = 1/(Re\,Pr)$, $\kappa_S = 1/(Re\,Sc)$. Its roots have opposite signs ($a < 0$, $\Lambda \le 0$, $G_z > 0$ for positive salinity in the water): the physical root is the non-negative one. Then $s_b = z - \sigma$, $\theta_b = \Lambda s_b + \theta_f$ and $m^* = A + Bz$. The local problem is solved in physical space (dealiased grid) and the results are transformed back to Fourier space.

Velocity. The scalars are solved before the velocity in each substep, and the velocity solve uses $v(0,x) = m^*(x)$, $u(0,x) = 0$ (`melt_blowing`), an inhomogeneous wall value of v handled by the same superposition as the other wall conditions. In 2D the mean ($k_x = 0$) part of continuity gives $\partial\bar{v}/\partial y = 0$, so the mean meltwater velocity $\bar{v} = \overline{m^*}$ is the same at every y and leaves the domain at $y = L_y$: the whole water column moves away from the ice at $\overline{m^*}$, as the far field does in the frame of the interface.

Output. `log.dat` gives the x-averages of m^*, θ_b and s_b and the number of coupling iterations; `melt_NNNNNNN.dat` gives $m^*(x)$, $\theta_b(x)$, $s_b(x)$ at every field output.

Validation. Steady melting has exact solutions in the frame of the interface. With the meltwater velocity $V = m^*$ the profiles are exponential: $\theta = \theta_b + (\theta_t - \theta_b)(e^{Vy/\kappa_T} - 1)/(e^{VL_y/\kappa_T} - 1)$ and, since the total salt flux vanishes, $z = z_b\, e^{Vy/\kappa_S}$. The heat condition then gives $1 = St\,(\theta_t - \theta_b)/(e^{VL_y/\kappa_T} - 1)$; without salt, $V = (\kappa_T/L_y)\ln(1 + St)$. The code reproduces these to the discretisation error (relative 6×10^{-9} for m^* in the Stefan case; 1.4×10^{-8} and 1.6×10^{-9} on 65 and 129 points for the three-equation case).

Time stepping. Crank–Nicolson (θ-method with $\theta = 1/2$) is not L-stable: after an abrupt change at the interface, e.g. from an initial state that does not satisfy the melt condition, the wall gradient and hence m^* oscillate from step to step, and the oscillation decays slowly on fine grids. The implicit Euler method (`scal_theta = 1`) damps it and can be used to start a run.

6 Comparison with Wells & Worster (2011)

Wells & Worster (2011) write the interface as $y = a(t)$, with the solid at $y < a$ and the liquid at $y > a$, so that $\dot a = \partial a/\partial t < 0$ for ablation; subscripts s (solid) and l (liquid), solute concentration C and solutal diffusivity D. With pressure neglected, their interface conditions are identical to (1)–(4) under the correspondence

$$\dot a = -m, \qquad \rho_s = \rho_i, \qquad \rho_l = \rho_w, \qquad c_l = c_w, \qquad \kappa_l = \kappa_T,$$
$$D = \kappa_S, \qquad C = S, \qquad C_s = S_i, \qquad T_i^{\mathrm{WW}} = T_b, \qquad C_i = S_b.$$

	Wells & Worster (2011)		this document
liquidus	$T_i = T_m - \Gamma\,(C_i - C_s)$	(2.1)	$T_b = \lambda_1 S_b + \lambda_2$: $\lambda_1 = -\Gamma$, $\lambda_2 = T_m + \Gamma C_s$
heat	$\rho_s L\dot a = \rho_s c_s \kappa_s T_y\|_{a^-} - \rho_l c_l \kappa_l T_y\|_{a^+}$	(2.2)	(2) with $\dot a = -m$
salt	$\rho_s(C_i - C_s)\,\dot a = -\rho_l D\, C_y\|_{a^+}$	(2.3)	(3) with $\dot a = -m$
velocity	$v(0) = -(\rho_s/\rho_l)\,\dot a$	(3.5a)	(4): $v(0) = (\rho_i/\rho_w)\,m$

Remarks:

- Liquidus: Wells & Worster use $T_m = 0\,^\circ\mathrm{C}$ and $\Gamma \approx 0.06\,^\circ\mathrm{C\,psu^{-1}}$ with $C_s \approx 0$, i.e. $\lambda_1 \approx -0.06\,^\circ\mathrm{C\,psu^{-1}}$ and $\lambda_2 = 0$. Holland & Jenkins (1999) fit $\lambda_1 \approx -0.057\,^\circ\mathrm{C\,psu^{-1}}$ and $\lambda_2 \approx 0.083\,^\circ\mathrm{C}$ over the oceanic salinity range; the difference in T_b is below $0.1\,^\circ\mathrm{C}$.
- Heat: Wells & Worster show (their appendix) that the conductive heat flux into the solid is typically small and neglect it, i.e. $q_i = 0$ in (11).
- Velocity: in the interface-fixed frame the wall-normal velocity at $y = 0$ equals the meltwater volume flux $(\rho_i/\rho_w)\,m$ and is positive (directed into the water). It equals the ablation rate m itself only if $\rho_i = \rho_w$.
- Their ablation rate, (3.15), $\dot a = -(\rho_l/\rho_s)\,v(0)$, is (4) solved for m. They also note that the fluid flow, and $v(0)$ itself, do not depend on ρ_s/ρ_l; only the conversion from $v(0)$ to the ablation rate does. This agrees with (11), which determines $m^* = v(0)/U$ without ρ_i/ρ_w.
- Buoyancy: Wells & Worster keep only the compositional (salt) buoyancy, $\rho = \rho_\infty + \rho_l\beta(C - C_\infty)$; `solver2d` has both, through `ri(1)` and `ri(2)` (set `ri(1)` $= 0$ to reproduce their assumption).

References

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